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Ingeniería en Ciencias y Sistemas
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Redes de Computadoras 2

Banco 5

Funciones, procedimientos y triggers

Creación de tablas y capturas

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1. Router

```
Banco5-Router#
Banco5-Router#enable
Banco5-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Banco5-Router(config)#interface e1/2
Banco5-Router(config-if)#ip address 172.20.5.153 255.255.255.252
Banco5-Router(config-if)#no shutdown
Banco5-Router(config-if)#end
Banco5-Router#
*Aug 30 19:46:35.823: %LINK-3-UPDOWN: Interface Ethernet1/2, changed state to up
Banco5-Router#
*Aug 30 19:46:36.647: %SYS-5-CONFIG_I: Configured from console by console
*Aug 30 19:46:36.823: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/2, changed state to up
Banco5-Router#write memory
Building configuration...
[OK]
Banco5-Router#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down
Ethernet1/0	unassigned	YES	unset	administratively down	down
Ethernet1/1	unassigned	YES	unset	administratively down	down
Ethernet1/2	172.20.5.153	YES	manual	up	up
Ethernet1/3	unassigned	YES	unset	administratively down	down

```
Banco5-Router#
```

2. Switch

2.1. VLANS

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name CAJAS
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name RRHH
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name SISTEMAS
Switch(config-vlan)#vlan 40
Switch(config-vlan)#name ADMINISTRACION
Switch(config-vlan)#vlan 50
Switch(config-vlan)#name SERVIDORES
Switch(config-vlan)#exit
```

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2, Gi0/3 Gi1/0, Gi1/1, Gi1/2, Gi1/3 Gi2/0, Gi2/1, Gi2/2, Gi2/3 Gi3/0, Gi3/1, Gi3/2, Gi3/3
10	CAJAS	active	
20	RRHH	active	
30	SISTEMAS	active	
40	ADMINISTRACION	active	
50	SERVIDORES	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
Switch#write memory
```

3. Configuración vlans

```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#ip routing
Switch(config)#interface vlan 10
Switch(config-if)#
*Aug 31 00:05:22.796: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to down
Switch(config-if)#ip address 172.20.5.1 255.255.255.224
Switch(config-if)#no shutdown
Switch(config-if)#
Switch(config-if)#
*Aug 31 00:06:14.706: %LINK-3-UPDOWN: Interface Vlan10, changed state to down
% Incomplete command.

Switch(config)#interface vlan 20
Switch(config-if)#ip address
*Aug 31 00:08:12.911: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to
Switch#
*Aug 31 00:08:19.425: %SYS-5-CONFIG_I: Configured from console by console
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlan 20
Switch(config-if)#ip address 172.20.5.33 255.255.255.224
Switch(config-if)#no shutdown
Switch(config-if)#
*Aug 31 00:09:27.104: %LINK-3-UPDOWN: Interface Vlan20, changed state to down
Switch(config-if)#interface vlan 30
Switch(config-if)#
*Aug 31 00:09:42.950: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to down
Switch(config-if)#ip address 172.20.5.65 255.255.255.224
Switch(config-if)#interface vlan 40
Switch(config-if)#
*Aug 31 00:15:10.492: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan40, changed state to down
Switch(config-if)#ip address 172.20.5.97 255.255.255.224
Switch(config-if)#no shutdown
Switch(config-if)#
*Aug 31 00:16:29.733: %LINK-3-UPDOWN: Interface Vlan40, changed state to down
Switch(config-if)#interface vlan 30
Switch(config-if)#no shutdown
Switch(config-if)#interface
*Aug 31 00:16:46.011: %LINK-3-UPDOWN: Interface Vlan30, changed state to
Switch(config-if)#interface vlan 50
Switch(config-if)#ip address
*Aug 31 00:16:57.259: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan50, changed state to
Switch(config-if)#ip address 172.20.5.129 255.255.255.240
Switch(config-if)#no shutdown
Switch(config-if)#
*Aug 31 00:17:45.799: %LINK-3-UPDOWN: Interface Vlan50, changed state to down
```

3.2. Configuración puerto firewall no trunk

```
Switch(config-if)#interface g0/0
Switch(config-if)#no switchport
Switch(config-if)#ip address
*Aug 31 01:00:49.057: %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down
*Aug 31 01:00:50.057: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed stat
Switch(config-if)#ip address 172.20.5.158 255.255.255.252
Switch(config-if)#no shutdown
Switch(config-if)#
*Aug 31 01:01:11.638: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Aug 31 01:01:12.638: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Switch(config-if)#exit
Switch(config)#
```

3.3. Salida por Firewall de vlans

```
Switch(config)#ip route 0.0.0.0 0.0.0.0 172.20.5.157
Switch(config)#
```

3.4. Firewall

```
# ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue sta
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
19: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdis
link/ether 02:42:22:a3:d1:00 brd ff:ff:ff:ff:ff:ff
20: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdis
link/ether 02:42:22:a3:d1:01 brd ff:ff:ff:ff:ff:ff
/ # ip addr add 172.20.5.154/30 dev eth0
/ # ip link set eth0 up
/ # ip addr add 172.20.5.157/30 dev eth1
/ # ip link set eth1 up
ip: ioctl 0x8913 failed: No such device
/ # ip link set eth1 up
/ #
```

4. IP FORWARDING EN FIREWALL

```
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
    address 172.20.5.154
    netmask 255.255.255.252

auto eth1
iface eth1 inet static
    address 172.20.5.157
    netmask 255.255.255.252

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#    hostname AlpineLinux-1
~
~
~
~
~
```

```
# content of this file will override /etc/sysctl.d/*
net.ipv4.ip_forward = 1
~
~
```

4.1. Configuración de rutas estáticas para llevar a vlan desde router

```
Banco5-Router#
Banco5-Router#enable
Banco5-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Banco5-Router(config)#ip route 172.20.5.0 255.255.255.0 172.20.5.154
Banco5-Router(config)#end
Banco5-Router#w
*Aug 31 01:37:20.963: %SYS-5-CONFIG_I: Configured from console by console
Banco5-Router#write memory
Building configuration...
[OK]
Banco5-Router#
```

4.2. Política para banco 5

Origen	Destino	¿Permitido?	Justificación
Cajas (VLAN10)	Servidores (VLAN50)	Sí, solo puertos de la app bancaria	Cajas solo necesita hablar con el servicio bancario
Cajas (VLAN10)	Internet	No	Terminal crítica, sin necesidad de salir a Internet
Cajas (VLAN10)	Otras VLAN (RRHH/Sistemas/Admin)	No	Aislamiento total entre áreas de negocio
RRHH (VLAN20)	Servidores (VLAN50)	Sí, solo su servicio de archivos	Acceso mínimo necesario
RRHH (VLAN20)	Internet	Sí, limitado (control de dominios, siguiente etapa)	Uso normal de oficina
Sistemas (VLAN30)	Servidores (VLAN50)	Sí, todos los puertos de gestión	Soporte técnico necesita administrar servidores
Sistemas (VLAN30)	Internet	Sí	Uso normal + actualizaciones
Administración (VLAN40)	Servidores (VLAN50)	Sí, acceso a app bancaria y reportes	Gerencia consulta información
Administración (VLAN40)	Internet	Sí	Uso normal
Servidores (VLAN50)	Cualquiera (saliente)	No, salvo respuestas a conexiones ya establecidas	Los servidores no deben iniciar conexiones hacia clientes
Cualquiera	Router (administración SSH/Telnet)	Solo desde Sistemas	Solo IT administra la red

5. FORWARD DE FIREWALL

```
#!/bin/sh
# Limpiar reglas existentes
iptables -F
iptables -X

# Politicas por defecto: Principio de minimo privilegio
iptables -P INPUT DROP
iptables -P FORWARD DROP
iptables -P OUTPUT ACCEPT

#Loopback siempre permitido
iptables -A INPUT -i lo -j ACCEPT

# Trafico de retorno (conexiones ya establecidas)
iptables -A INPUT -m conntrack --ctstate ESTABLISHED, RELATED -j ACCEPT
iptables -A FORWARD -m conntrack --ctstate ESTABLISHED, RELATED -j ACCEPT

#Administracion del propio firewall: solo SSH desde VLAN Sistemas
iptables -A INPUT -p tcp -s 172.20.5.64/27 --dport 22 -j ACCEPT

# --- Cajas (VLAN10) -> Servidor1 app bancaria, puerto 80 ---
iptables -A FORWARD -s 172.20.5.0/27 -d 172.20.5.130 -p tcp --dport 80 -j ACCEPT

# --- RRHH (VLAN20) -> Servidor2 archivos, puerto 22 ---
iptables -A FORWARD -s 172.20.5.32/27 -d 172.20.5.131 -p tcp --dport 22 -j ACCEPT

# --- RRHH (VLAN20) -> Internet (se afinara con control de dominios) ---
iptables -A FORWARD -s 172.20.5.32/27 -o eth0 -j ACCEPT

# --- Sistemas (VLAN30) -> gestion de ambos servidores (SSH 22, y HTTP 80 para pruebas) ---
iptables -A FORWARD -s 172.20.5.64/27 -d 172.20.5.130 -p tcp -m multiport --dports 22,
iptables -A FORWARD -s 172.20.5.64/27 -d 172.20.5.131 -p tcp --dport 22 -j ACCEPT

# --- Sistemas (VLAN30) -> Internet ---
iptables -A FORWARD -s 172.20.5.64/27 -o eth0 -j ACCEPT

# --- Administracion (VLAN40) -> Servidor1 (reportes, puerto80) ---
iptables -A FORWARD -s 172.20.5.96/27 -d 172.20.5.130 -p tcp --dport 80 -j ACCEPT

# --- Administracion (VLAN40) -> Internet ---
iptables -A FORWARD -s 172.20.5.96/27 -o eth0 -j ACCEPT

#Todo lo demas queda bloqueado por la politica por defecto:
# - Cajas sin Internet
# - Servidores sin salida propia (solo responden, via conntrack)
# - Comunicacion directa entre VLAN (Cajas<-> RRHH, etc.) bloqueada

~
~
~
```

```
cat << 'EOF' > /etc/nginx/http.d/default.conf
```

```
server {
    listen 80 default_server;
```

```

listen [::]:80 default_server;

root /var/www/localhost/htdocs;
index index.html;

location / {
    try_files $uri $uri/ =404;
}
}
EOF

```

5.1. Servidor 1

```

# cat /etc/nginx/http.d/default.conf
# This is a default site configuration which will simply return
# chance access to any other virtualhost.

server {
    listen 80 default_server;
    listen [::]:80 default_server;

    root /var/www/localhost/htdocs;
    index index.html;

    # Everything is a 404
    location / {
        return 404;
    }

    # You may need this to prevent return 404 recursion.
    location = /404.html {
        internal;
    }
}
/ # vi /etc/nginx/http.d/default.conf
/ # pkill nginx
/ # nginx
/ # curl http://localhost
<h1>Banco 5 - Servicio de Aplicacion Bancaria</h1>

```

redes1234

5.2. Servidor 2

```
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
    address 172.20.5.131
    netmask 255.255.255.240
    gateway 172.20.5.129
#    up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#    hostname AlpineLinux-1
~
~
~
~
~
```

5.3. Servidor 1

```
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
    address 172.20.5.130
    netmask 255.255.255.240
    gateway 172.20.5.129
#    up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#    hostname AlpineLinux-1
~
~
~
~
~
~
~
```

Configuracion vlans

VLAN Name	Status	Ports
1 default	active	Gi0/3, Gi1/0, Gi1/1, Gi1/2 Gi1/3, Gi2/0, Gi2/1, Gi2/2 Gi2/3, Gi3/0, Gi3/1, Gi3/2 Gi3/3
10 CAJAS	active	
20 RRHH	active	
30 SISTEMAS	active	
40 ADMINISTRACION	active	
50 SERVIDORES	active	Gi0/1, Gi0/2
1002 fddi-default	act/unsup	
1003 token-ring-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trnet-default	act/unsup	

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface GigabitEthernet1/0
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#no shutdown
Switch(config-if)#interface GigabitEthernet1/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchpo
*Aug 31 06:52:11.531: %LINK-3-UPDOWN: Interface Vlan10, changed state to up
*Aug 31 06:52:12.533: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed sta
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shutdown
Switch(config-if)#interface GigabitEthernet1/2
Switch(config-if)#switchport mode access
Switch(config-if)#
*Aug 31 06:53:02.865: %LINK-3-UPDOWN: Interface Vlan20, changed state to up
*Aug 31 06:53:03.865: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
Switch(config-if)#switchport access vlan 30
Switch(config-if)#no shutdown
Switch(config-if)#interface GigabitEthernet1/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#no shutdown
Switch(config-if)#end
Switch#
*Aug 31 06:53:48.456: %SYS-5-CONFIG_I: Configured from console by console
*Aug 31 06:53:49.719: %LINK-3-UPDOWN: Interface Vlan30, changed state to up
*Aug 31 06:53:50.719: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up
Switch#write memory
Building configuration...
Compressed configuration from 3901 bytes to 2062 bytes[OK]
Switch#
*Aug 31 06:53:58.919: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Aug 31 06:53:59.572: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Switch#

```

Cajas funcionando

```
Trying ::1...
Connection failed: Conexión rehusada
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
/ # curl 172.20.5.130
curl: (7) Failed to connect to 172.20.5.130:80 after 0 ms: Could not connect to server
/ # curl 172.20.5.130
curl: (7) Failed to connect to 172.20.5.130:80 after 0 ms: Could not connect to server
/ # curl http://172.20.5.130
curl: (7) Failed to connect to 172.20.5.130:80 after 0 ms: Could not connect to server
/ # ip addr add 172.20.5.130/28 dev eth0
/ # ip route add default via 172.20.5.129
/ # curl http://172.20.5.129
curl: (7) Failed to connect to 172.20.5.129:80 after 3063 ms: Could not connect to server
/ # curl http://172.20.5.130
curl: (7) Failed to connect to 172.20.5.130:80 after 0 ms: Could not connect to server
/ # curl http://172.20.5.130
curl: (7) Failed to connect to 172.20.5.130:80 after 0 ms: Could not connect to server
/ # ip addr flush dev eth0
/ # ip addr add 172.20.5.10/27 dev eth0
ip: either "local" is duplicate, or "dev" is garbage
/ # ip addr add 172.20.5.10/27 dev eth0
/ # ip route add default via 172.20.5.129
ip: RTNETLINK answers: Network unreachable
/ # ip route add default via 172.20.5.1
/ # ip addr show eth0
25: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN qlen 1000
    link/ether 02:42:dd:11:dc:00 brd ff:ff:ff:ff:ff:ff
    inet 172.20.5.10/27 scope global eth0
        valid_lft forever preferred_lft forever
/ # curl http://172.20.5.130
<h1>Banco 5 - Servicio de Aplicacion Bancaria</h1>
/ # █
```

SERVIDOR 2

```
Trying ::1...
Connection failed: Conexión rehusada
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
/ # udhcpc -i eth0
udhcpc: ioctl 0x8933 failed: No such device
/ # udhcpc -i eth0
udhcpc: started, v1.37.0
udhcpc: broadcasting discover
udhcpc: broadcasting select for 192.168.122.161, server 192.168.122.1
udhcpc: lease of 192.168.122.161 obtained from 192.168.122.1, lease time 3600
/ # apk update
v3.24.1-474-gd70c963d846 [https://dl-cdn.alpinelinux.org/alpine/v3.24/main]
v3.24.1-472-g86918b56b21 [https://dl-cdn.alpinelinux.org/alpine/v3.24/community]
OK: 28645 distinct packages available
/ # apk add openssh
(1/6) Installing bridge (1.5-r5)
(2/6) Installing ifupdown-ng (0.13.0-r0)
(3/6) Installing openrc-user (0.63.2-r0)
(4/6) Installing libcap2 (2.78-r0)
(5/6) Installing openrc (0.63.2-r0)
Executing openrc-0.63.2-r0.post-install
(6/6) Installing openssh-server-common-openrc (10.3_p1-r1)
Executing busybox-1.37.0-r31.trigger
OK: 15.5 MiB in 32 packages
/ # adduser rrhh_user
adduser: user 'rrhh_user' in use
/ # passwd rrhh_user
Changing password for rrhh_user
New password:
Retype password:
passwd: password for rrhh_user changed by root
/ # rc-s
rc-service rc-sstat rc-status
/ # rc-s
rc-service rc-sstat rc-status
/ # rc-service sshd start
* sshd: failed to acquire lock: Bad file descriptor
/ # apk add openssh openssh-server-pam openrc
(1/4) Installing skalibs-libs (2.15.0.0-r0)
(2/4) Installing utmps-libs (0.1.3.3-r0)
(3/4) Installing linux-pam (1.7.1-r2)
(4/4) Installing openssh-server-pam (10.3_p1-r1)
Executing busybox-1.37.0-r31.trigger
OK: 18.3 MiB in 36 packages
/ # rc-service sshd start
* sshd: failed to acquire lock: Bad file descriptor
/ # mkdir -p /run/openrc
/ # touch /run/openrc/softlevel
/ # rc-service sshd start
* Caching service dependencies ...
Service 'hwdrivers' needs non existent service 'dev'
Service 'machine-id' needs non existent service 'dev'
ssh-keygen: generating new host keys: RSA ECDSA ED25519
* Starting sshd ...
/ # rc-update add sshd default
* service sshd added to runlevel default
/ # rc-service sshd status
* status: started
/ # █
```

```
Trying ::1...
Connection failed: Conexión rehusada
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
/ # adduser rrhh_user
Changing password for rrhh_user
New password:
Retype password:
passwd: password for rrhh_user changed by root
/ # mkdir -p /run/openrc
/ # touch /run/openrc/softlevel
/ # rc-service sshd start
* Caching service dependencies ...
Service 'hwdrivers' needs non existent service 'dev'
Service 'machine-id' needs non existent service 'dev'
* Starting sshd ...
/ # █
```

6. Conexión de RRHH a Servidor 2

```

Trying ::1...
Connection failed: Conexión rehusada
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
/ # ssh rrhh_user@172.20.5.131\
>
^C
/ # ssh rrhh_user@172.20.5.131
^C
/ # ping -c 3 172.20.5.131
PING 172.20.5.131 (172.20.5.131): 56 data bytes
^C
--- 172.20.5.131 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
/ # ip addr add 172.20.5.40/27 dev eth9
ip: can't find device 'eth9'
/ # ip addr add 172.20.5.40/27 dev eth0
ip: RTNETLINK answers: File exists
/ # ip link set eth0 up
/ # ip route add default via 172.20.5.33
ip: RTNETLINK answers: File exists
/ # ssh rrhh_user@172.20.5.131
^C
/ # ssh rrhh_user@172.20.5.131
^C
/ # ip addr flush dev eth0
/ # ip addr add 172.20.5.40/27 dev eth0
/ # ip route add default via 172.20.5.33
/ # ping -c 3 172.20.5.131
PING 172.20.5.131 (172.20.5.131): 56 data bytes

^C
--- 172.20.5.131 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
/ # ping -c 3 172.20.5.131
PING 172.20.5.131 (172.20.5.131): 56 data bytes
64 bytes from 172.20.5.131: seq=0 ttl=63 time=9.640 ms
64 bytes from 172.20.5.131: seq=1 ttl=63 time=3.072 ms
64 bytes from 172.20.5.131: seq=2 ttl=63 time=1.518 ms

--- 172.20.5.131 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.518/4.743/9.640 ms
/ # ssh rrhh_user@172.20.5.131
The authenticity of host '172.20.5.131 (172.20.5.131)' can't be established.
ED25519 key fingerprint is: SHA256:iwNzoMaU0KB9Z6Vx2/7m13dFUXax6Y2YjIdvo20i2Dg
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
Host key verification failed.
/ # ssh rrhh_user@172.20.5.131
The authenticity of host '172.20.5.131 (172.20.5.131)' can't be established.
ED25519 key fingerprint is: SHA256:iwNzoMaU0KB9Z6Vx2/7m13dFUXax6Y2YjIdvo20i2Dg
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? prueba1234
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '172.20.5.131' (ED25519) to the list of known hosts.
rrhh_user@172.20.5.131's password:
Welcome to Alpine!

The Alpine Wiki contains a large amount of how-to guides and general
information about administrating Alpine systems.
See <https://wiki.alpinelinux.org/>.

You can setup the system with the command: setup-alpine

You may change this message by editing /etc/motd.

Banco5-Servidor2:~$ █

```

7. PROCEDIMIENTO PARTE D

7.1. Procedimiento

The screenshot displays a database IDE with two main panels. The top panel shows the SQL script for creating a procedure, and the bottom panel shows the execution results and a data grid.

Script-21 x

```
•CREATE OR REPLACE PROCEDURE extender_prestamo(p_id INT, p_dias_extra INT)
LANGUAGE plpgsql AS $$
BEGIN
    UPDATE prestamo
    SET dias = dias + p_dias_extra
    WHERE id = p_id;
END;
$$;
```

Estadísticas 1 x

Name	Value
Updated Rows	0
Execute time	0.006s
Start time	Fri Aug 28 23:01:47 CST 2026
Finish time	Fri Aug 28 23:01:47 CST 2026
Query	CREATE OR REPLACE PROCEDURE extender_prestamo(p_id INT, p_dias_extra INT) LANGUAGE plpgsql AS \$\$ BEGIN UPDATE prestamo SET dias = dias + p_dias_extra WHERE id = p_id; END;

***<biblioteca> Script-21 x**

```
CALL extender_prestamo(1, 3);

SELECT * FROM prestamo WHERE id = 1;
```

prestamo 1 x

	123 id	123 idsocio	123 idlibro	123 dias	fecha
1	1	1	1	7	2026-08-01

8. TRIGGER PARTE E

8.1. Tabla Log

The screenshot displays a database IDE with a SQL script for creating a table.

***<biblioteca> Script-21 x**

```
•CREATE TABLE libro_log (
    id SERIAL PRIMARY KEY,
    libro_titulo VARCHAR(150),
    fecha TIMESTAMP DEFAULT NOW())
```

8.2. Funcion del trigger

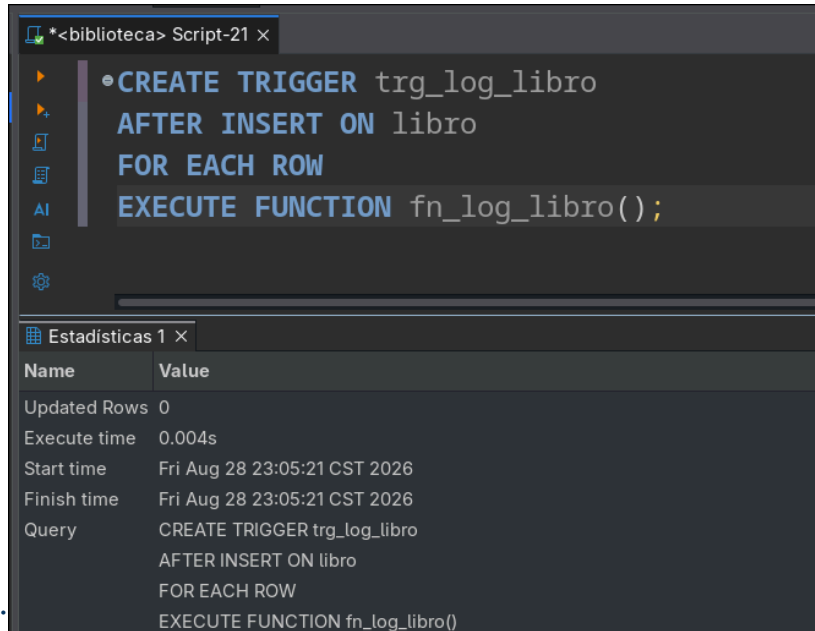
The screenshot shows a database IDE with a script editor and a statistics panel. The script editor contains the following PL/SQL code:

```
CREATE OR REPLACE FUNCTION fn_log_libro()  
RETURNS TRIGGER AS $$  
BEGIN  
    INSERT INTO libro_log(libro_titulo, fecha)  
    VALUES (NEW.titulo, NOW());  
    RETURN NEW;  
END;  
$$ LANGUAGE plpgsql;
```

The statistics panel, titled "Estadísticas 1", displays the following information:

Name	Value
Updated Rows	0
Execute time	0.003s
Start time	Fri Aug 28 23:04:32 CST 2026
Finish time	Fri Aug 28 23:04:32 CST 2026
Query	CREATE OR REPLACE FUNCTION fn_log_libro() RETURNS TRIGGER AS \$\$ BEGIN INSERT INTO libro_log(libro_titulo, fecha) VALUES (NEW.titulo, NOW()); RETURN NEW; END; \$\$ LANGUAGE plpgsql

8.3. Trigger despues del insert

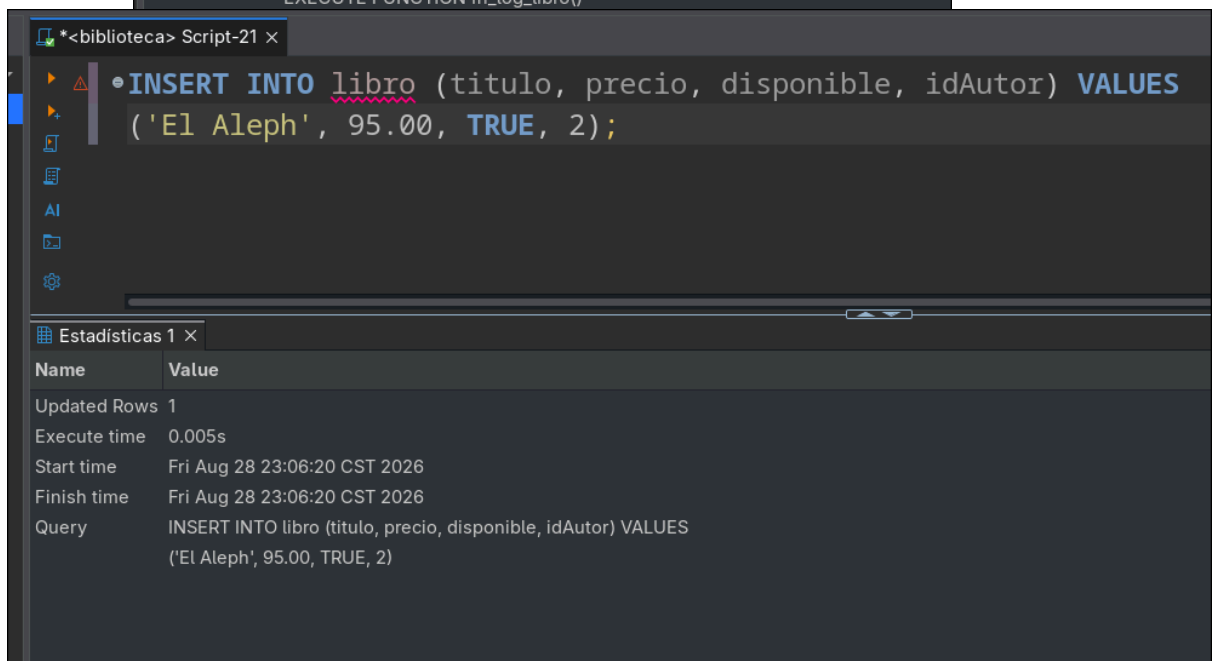


The screenshot shows a SQL script editor with a dark theme. The script defines a trigger named `trg_log_libro` that fires after an insert on the `libro` table. The trigger body calls the function `fn_log_libro()`. Below the script, a statistics window titled "Estadísticas 1" is open, displaying the execution details of the trigger creation.

```
CREATE TRIGGER trg_log_libro
AFTER INSERT ON libro
FOR EACH ROW
EXECUTE FUNCTION fn_log_libro();
```

Name	Value
Updated Rows	0
Execute time	0.004s
Start time	Fri Aug 28 23:05:21 CST 2026
Finish time	Fri Aug 28 23:05:21 CST 2026
Query	CREATE TRIGGER trg_log_libro AFTER INSERT ON libro FOR EACH ROW EXECUTE FUNCTION fn_log_libro()

8.4.

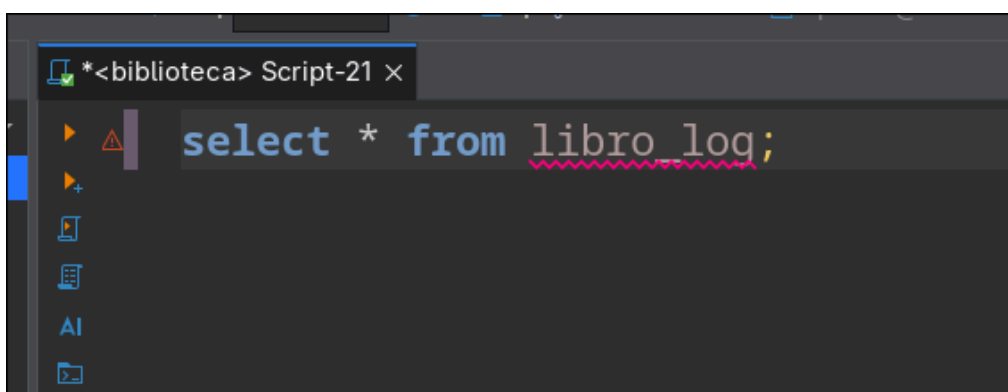


The screenshot shows a SQL script editor with a dark theme. The script contains an `INSERT INTO` statement for the `libro` table, inserting a record with the title 'El Aleph', a price of 95.00, and a status of TRUE, with an author ID of 2. Below the script, a statistics window titled "Estadísticas 1" is open, displaying the execution details of the insert statement.

```
INSERT INTO libro (titulo, precio, disponible, idAutor) VALUES
('El Aleph', 95.00, TRUE, 2);
```

Name	Value
Updated Rows	1
Execute time	0.005s
Start time	Fri Aug 28 23:06:20 CST 2026
Finish time	Fri Aug 28 23:06:20 CST 2026
Query	INSERT INTO libro (titulo, precio, disponible, idAutor) VALUES ('El Aleph', 95.00, TRUE, 2)

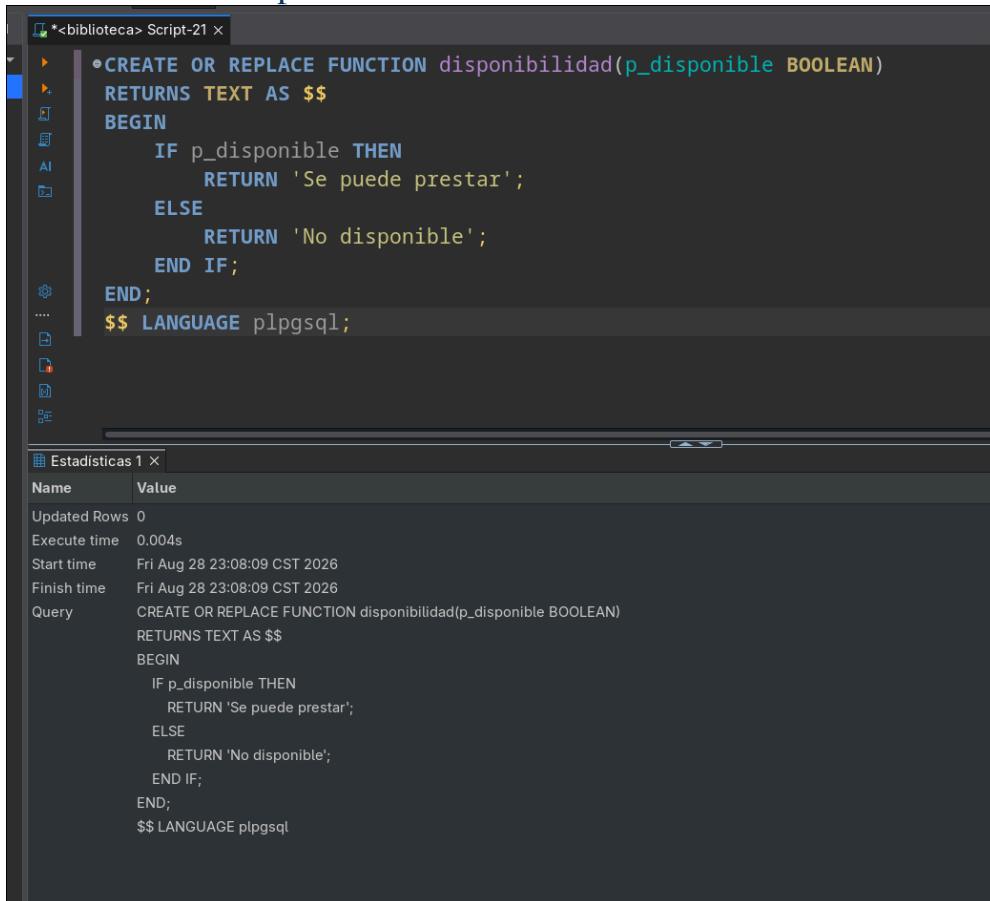
8.5. Select del libro_log



The screenshot shows a SQL script editor with a dark theme. The script contains a `SELECT` statement that retrieves all data from the `libro_log` table.

```
select * from libro_log;
```

8.6. Funcion disponibilidad



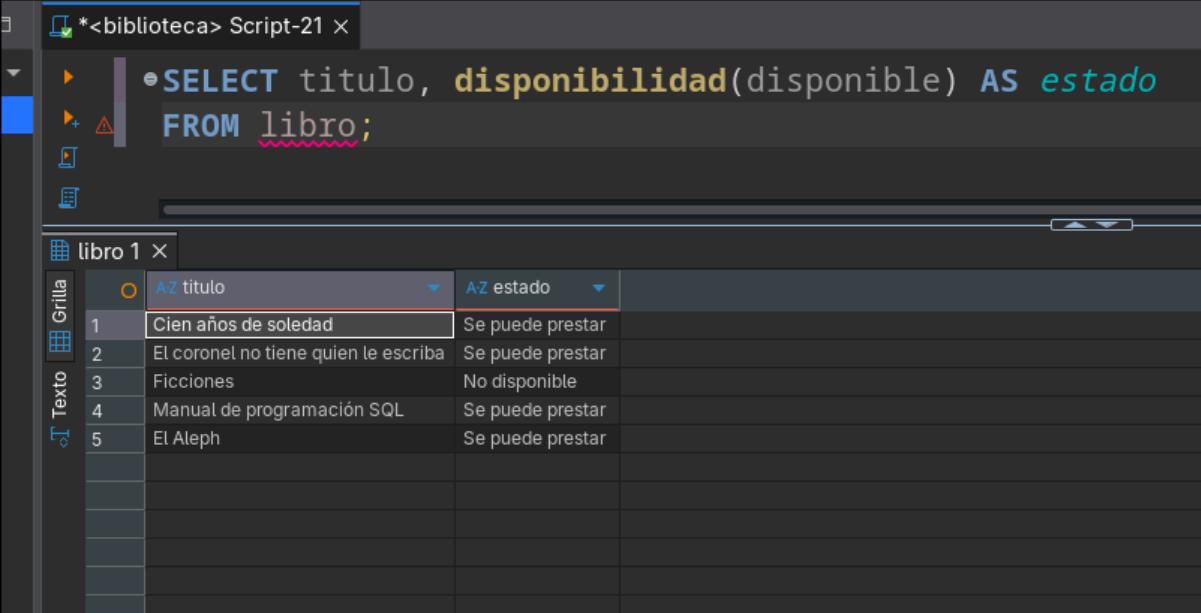
The screenshot displays a PostgreSQL IDE window titled "<biblioteca> Script-21 x". The main editor contains the following SQL code:

```
•CREATE OR REPLACE FUNCTION disponibilidad(p_disponible BOOLEAN)
RETURNS TEXT AS $$
BEGIN
    IF p_disponible THEN
        RETURN 'Se puede prestar';
    ELSE
        RETURN 'No disponible';
    END IF;
END;
$$ LANGUAGE plpgsql;
```

Below the editor, the "Estadísticas 1 x" (Statistics) tab is active, showing the execution details of the query. The statistics are as follows:

Name	Value
Updated Rows	0
Execute time	0.004s
Start time	Fri Aug 28 23:08:09 CST 2026
Finish time	Fri Aug 28 23:08:09 CST 2026
Query	CREATE OR REPLACE FUNCTION disponibilidad(p_disponible BOOLEAN) RETURNS TEXT AS \$\$ BEGIN IF p_disponible THEN RETURN 'Se puede prestar'; ELSE RETURN 'No disponible'; END IF; END; \$\$ LANGUAGE plpgsql

8.7. Uso de funcion disponibilidad sobre libro



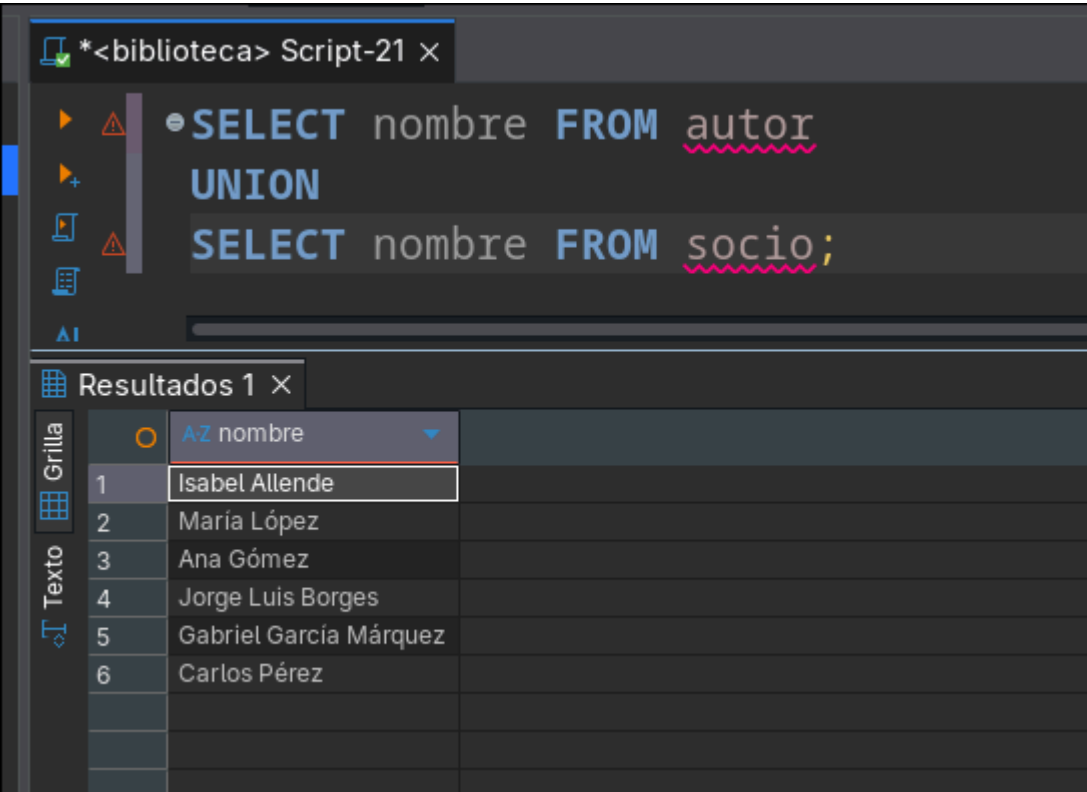
The screenshot shows a database script editor with a query and its results. The query is:

```
SELECT titulo, disponibilidad(disponible) AS estado
FROM libro;
```

The results are displayed in a table with the following data:

	AZ titulo	AZ estado
1	Cien años de soledad	Se puede prestar
2	El coronel no tiene quien le escriba	Se puede prestar
3	Ficciones	No disponible
4	Manual de programación SQL	Se puede prestar
5	El Aleph	Se puede prestar

8.8. UNION: Combinar nombres de autores y de socios en un asola linea



The screenshot shows a database script editor with a query using UNION and its results. The query is:

```
SELECT nombre FROM autor
UNION
SELECT nombre FROM socio;
```

The results are displayed in a table with the following data:

	AZ nombre
1	Isabel Allende
2	María López
3	Ana Gómez
4	Jorge Luis Borges
5	Gabriel García Márquez
6	Carlos Pérez

CAJAS

```
# Static config for eth0\  
auto lo  
iface lo inet loopback  
  
auto eth0  
iface eth0 inet static  
    address 172.20.5.10  
    netmask 255.255.255.224  
    gateway 172.20.5.1  
#    up echo nameserver 192.168.0.1 > /etc/resolv.conf  
  
# DHCP config for eth0  
#auto eth0  
#iface eth0 inet dhcp  
#    hostname AlpineLinux-1
```

Refresh

Save

Cancel

RRHH

```
#  
# This is a sample network config, please uncomment lines to  
# configure the network  
#  
# Uncomment this line to load custom interface files  
# source /etc/network/interfaces.d/*  
auto lo  
iface lo inet loopback  
  
# Static config for eth0  
auto eth0  
iface eth0 inet static  
    address 172.20.5.40  
    netmask 255.255.255.224  
    gateway 172.20.5.33  
#    up echo nameserver 192.168.0.1 > /etc/resolv.conf  
  
# DHCP config for eth0  
#auto eth0  
#iface eth0 inet dhcp  
#    hostname AlpineLinux-1
```

Refresh

 Save

 Cancel

SWITCH NUEVO

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#no ip routing
Switch(config)#ip access-list extended CAJAS_ACL
Switch(config-ext-nacl)#172.20.5.0 0.0.0.31 host 172.20.5.130 eq 80
Switch(config-ext-nacl)#deny ip 172.20.5.0 0.0.0.31 172.20.5.32 0.0.0.31
Switch(config-ext-nacl)#deny ip 172.20.5.0 0.0.0.31 172.20.5.64 0.0.0.31
Switch(config-ext-nacl)#deny ip 172.20.5.0 0.0.0.31 172.20.5.96 0.0.0.31
Switch(config-ext-nacl)#deny ip 172.20.5.0 0.0.0.31 172.20.5.128 0.0.0.31
Switch(config-ext-nacl)#permit ip any any
Switch(config-ext-nacl)#ip access-list extended RRHH_ACL
Switch(config-ext-nacl)#172.20.5.32 0.0.0.31 host 172.20.5.131 eq 22
Switch(config-ext-nacl)#deny ip 172.20.5.32 0.0.0.31 172.20.5.0 0.0.0.31
Switch(config-ext-nacl)#deny ip 172.20.5.32 0.0.0.31 172.20.5.64 0.0.0.31
Switch(config-ext-nacl)#deny ip 172.20.5.32 0.0.0.31 172.20.5.96 0.0.0.31
Switch(config-ext-nacl)#deny ip 172.20.5.32 0.0.0.31 172.20.5.128 0.0.0.15
Switch(config-ext-nacl)#permit ip any any
Switch(config-ext-nacl)#ip access-list extended SISTEMAS_ACL
Switch(config-ext-nacl)#172.20.5.64 0.0.0.31 host 172.20.5.130 eq 80
Switch(config-ext-nacl)#172.20.5.64 0.0.0.31 host 172.20.5.130 eq 22
Switch(config-ext-nacl)#172.20.5.64 0.0.0.31 host 172.20.5.131 eq 22
Switch(config-ext-nacl)#deny ip 172.20.5.64 0.0.0.31 172.20.5.0 0.0.0.31
Switch(config-ext-nacl)#deny ip 172.20.5.64 0.0.0.31 172.20.5.32 0.0.0.31
Switch(config-ext-nacl)#deny ip 172.20.5.64 0.0.0.31 172.20.5.96 0.0.0.31
Switch(config-ext-nacl)#permit ip any any
Switch(config-ext-nacl)#ip access-list extended ADMIN_ACL
Switch(config-ext-nacl)#172.20.5.96 0.0.0.31 host 172.20.5.130 eq 80
Switch(config-ext-nacl)#deny ip 172.20.5.96 0.0.0.31 172.20.5.0 0.0.0.31
Switch(config-ext-nacl)#deny ip 172.20.5.96 0.0.0.31 172.20.5.32 0.0.0.31
Switch(config-ext-nacl)#deny ip 172.20.5.96 0.0.0.31 172.20.5.64 0.0.0.31
Switch(config-ext-nacl)#deny ip 172.20.5.96 0.0.0.31 172.20.5.128 0.0.0.31
Switch(config-ext-nacl)#deny ip 172.20.5.96 0.0.0.31 172.20.5.128 0.0.0.15
Switch(config-ext-nacl)#permit ip any any
Switch(config-ext-nacl)#ip access-list extended SERVIDORES_ACL
Switch(config-ext-nacl)#$28 0.0.0.15 172.20.5.0 0.0.0.31 established
Switch(config-ext-nacl)#$28 0.0.0.15 172.20.5.32 0.0.0.31 established
Switch(config-ext-nacl)#$28 0.0.0.15 172.20.5.64 0.0.0.31 established
Switch(config-ext-nacl)#$28 0.0.0.15 172.20.5.96 0.0.0.31 established
Switch(config-ext-nacl)#permit ip 172.20.5.128 0.0.0.15 any
Switch(config-ext-nacl)#
```

SWITCH NUEVO TRUNK

```
Switch(config)#interface g0/0
Switch(config-if)#switchport mode trunk
Command rejected: Gi0/0 is not a switching port.
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,50
Command rejected: Gi0/0 is not a switching port.
Switch(config-if)#do show ip route
Default gateway is not set

Host          Gateway          Last Use    Total Uses  Interface
ICMP redirect cache is empty
Switch(config-if)#switchport
Switch(config-if)#swtch
*Sep  3 04:43:14.697: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to uc
*Sep  3 04:43:15.697: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,50
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,50
Switch(config-if)#no shutdown
Switch(config-if)#end
Switch#writ
*Sep  3 04:44:38.178: %SYS-5-CONFIG_I: Configured from console by con
Switch#write memory
Building configuration...
Compressed configuration from 5653 bytes to 2713 bytes[OK]
Switch#
*Sep  3 04:44:44.867: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Sep  3 04:44:45.519: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Switch#show interfaces trunk

Port      Mode          Encapsulation  Status      Native vlan
Gi0/0     on            802.1q         trunking    1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40,50

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40,50

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40,50
Switch#
```

FIREWALL

SERVIDOR 1

```
mkdir -p /var/www/localhost/cgi-bin/data
```

```
cat > /var/www/localhost/cgi-bin/data/cuentas.txt << 'EOF'
1001|Cajero Principal|5000.00
1002|Cliente A|1500.00
1003|Cliente B|800.00
EOF
```

```
cat > /var/www/localhost/cgi-bin/index.cgi << 'SCRIPT'
#!/bin/bash
echo "Content-type: text/html"
echo ""
DATA=/var/www/localhost/cgi-bin/data/cuentas.txt
cat << 'HTMLTOP'
<html>
<head><title>Banco 5 - Aplicacion Bancaria</title>
<style>
body { font-family: Arial, sans-serif; background:#f2f2f2; margin:0; padding:20px; }
.header { background:#003366; color:white; padding:15px; border-radius:6px; }
.card { background:white; border-radius:6px; padding:15px; margin-top:15px; box-shadow:0
1px 3px rgba(0,0,0,0.2); }
table { width:100%; border-collapse:collapse; }
th, td { text-align:left; padding:8px; border-bottom:1px solid #ddd; }
input { padding:6px; margin:4px 0; width:100%; box-sizing:border-box; }
button { background:#003366; color:white; border:none; padding:10px 16px;
border-radius:4px; cursor:pointer; }
</style>
</head>
<body>
<div class="header"><h1>Banco 5</h1><p>Servicio de Aplicacion Bancaria</p></div>
<div class="card">
<h2>Cuentas</h2>
<table>
<tr><th>No. Cuenta</th><th>Titular</th><th>Saldo</th></tr>
HTMLTOP
while IFS='|' read -r id nombre saldo; do
    printf "<tr><td>%s</td><td>%s</td><td>Q %.2f</td></tr>\n" "$id" "$nombre" "$saldo"
done < "$DATA"
cat << 'HTMLBOTTOM'
</table>
</div>
<div class="card">
<h2>Realizar Transferencia</h2>
<form method="post" action="/cgi-bin/transferir.cgi">
Cuenta origen:<br>
```

```

<input type="text" name="origen" placeholder="Ej. 1001"><br>
Cuenta destino:<br>
<input type="text" name="destino" placeholder="Ej. 1002"><br>
Monto:<br>
<input type="text" name="monto" placeholder="Ej. 100.00"><br><br>
<button type="submit">Transferir</button>
</form>
</div>
</body>
</html>
HTMLBOTTOM
SCRIPT

```

```

chmod +x /var/www/localhost/cgi-bin/index.cgi

```

```

cat > /var/www/localhost/cgi-bin/transferir.cgi << 'SCRIPT'
#!/bin/bash
echo "Content-type: text/html"
echo ""

```

```

DATA=/var/www/localhost/cgi-bin/data/cuentas.txt
TMP=$(mktemp)

```

```

read -n "$CONTENT_LENGTH" POST_DATA

```

```

urldecode() { local data="${1//+/ }"; printf '%b' "${data//%/\\x}"; }

```

```

ORIGEN=$(urldecode "$(echo "$POST_DATA" | tr '&' '\n' | grep '^origen=' | cut -d= -f2)")
DESTINO=$(urldecode "$(echo "$POST_DATA" | tr '&' '\n' | grep '^destino=' | cut -d= -f2)")
MONTO=$(urldecode "$(echo "$POST_DATA" | tr '&' '\n' | grep '^monto=' | cut -d= -f2)")

```

```

MENSAJE=""
SALDO_ORIGEN=""
ENCONTRADO_ORIGEN=0
ENCONTRADO_DESTINO=0

```

```

while IFS='|' read -r id nombre saldo; do
[ "$id" = "$ORIGEN" ] && { ENCONTRADO_ORIGEN=1; SALDO_ORIGEN=$saldo; }
[ "$id" = "$DESTINO" ] && ENCONTRADO_DESTINO=1
done < "$DATA"

```

```

if [ "$ENCONTRADO_ORIGEN" -eq 0 ] || [ "$ENCONTRADO_DESTINO" -eq 0 ]; then
MENSAJE="Error: cuenta origen o destino no existe."
elif ! echo "$MONTO" | grep -Eq '^[0-9]+(\.[0-9]+)?$'; then
MENSAJE="Error: monto invalido."
elif (( $(echo "$MONTO > $SALDO_ORIGEN" | bc -l) )); then
MENSAJE="Error: fondos insuficientes."
else

```

```

while IFS='|' read -r id nombre saldo; do
    if [ "$id" = "$ORIGEN" ]; then
        echo "$id|$nombre|$(echo "$saldo - $MONTO" | bc -l)" >> "$TMP"
    elif [ "$id" = "$DESTINO" ]; then
        echo "$id|$nombre|$(echo "$saldo + $MONTO" | bc -l)" >> "$TMP"
    else
        echo "$id|$nombre|$saldo" >> "$TMP"
    fi
done < "$DATA"
mv "$TMP" "$DATA"
MENSAJE="Transferencia realizada: Q $MONTO de $ORIGEN hacia $DESTINO."
fi

```

```

cat << HTML
<html>
<head><title>Resultado</title>
<style>body{font-family:Arial,sans-serif;background:#f2f2f2;padding:20px;}
.card{background:white;padding:15px;border-radius:6px;} a{color:#003366;}</style>
</head>
<body>
<div class="card">
<h2>${MENSAJE}</h2>
<p><a href="/cgi-bin/index.cgi">Volver al inicio</a></p>
</div>
</body>
</html>
HTML
SCRIPT

```

```

chmod +x /var/www/localhost/cgi-bin/transferir.cgi

```

```

cat > /etc/nginx/http.d/default.conf << 'EOF'
server {
    listen 80;
    server_name _;

    location / {
        return 302 /cgi-bin/index.cgi;
    }

    location /cgi-bin/ {
        root /var/www/localhost;
        gzip off;
        fastcgi_pass unix:/run/fcgiwrap.sock;
        fastcgi_index index.cgi;
        include fastcgi_params;
        fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
    }
}

```

```
}  
EOF
```

```
cat > /etc/network/interfaces << 'EOF'  
auto lo  
iface lo inet loopback  
  
auto eth0  
iface eth0 inet static  
    address 172.20.5.130  
    netmask 255.255.255.240  
    gateway 172.20.5.129  
EOF  
ifup -a  
rc-service nginx start  
spawn-fcgi -s /run/fcgiwrap.sock -M 766 -F 1 -- /usr/bin/fcgiwrap  
  
touch /etc/enable_nginx  
touch /etc/enable_fcgiwrap
```

TINY
<http://172.20.5.130>

```
config  
TINYCORE  
cat >> /opt/bootlocal.sh << 'EOF' ifconfig eth0 172.20.5.10 netmask 255.255.255.224 route  
add default gw 172.20.5.1 echo "nameserver 8.8.8.8" > /etc/resolv.conf EOF exit
```

PARA INTERBANCO 4

```
cat > /var/www/localhost/cgi-bin/blacklist_banco4.cgi << 'SCRIPT'  
#!/bin/bash  
echo "Content-type: text/html"  
echo ""
```

```
RESPUESTA=$(wget -qO- --timeout=5 http://10.0.0.13:8080/blacklist 2>/dev/null)
```

```
cat << 'HTMLTOP'  
<html>  
<head><title>Banco 5 - Blacklist de Banco 4</title>  
<style>  
body { font-family: Arial, sans-serif; background:#f2f2f2; margin:0; padding:20px; }  
.header { background:#003366; color:white; padding:15px; border-radius:6px; }  
.card { background:white; border-radius:6px; padding:15px; margin-top:15px; box-shadow:0  
1px 3px rgba(0,0,0,0.2); }  
ul { padding-left:20px; }
```

```

li { padding:4px 0; }
a { color:#003366; }
.error { color:#a00; font-weight:bold; }
</style>
</head>
<body>
<div class="header"><h1>Banco 5</h1><p>Consulta interbancaria: Blacklist de Banco
4</p></div>
<div class="card">
HTMLTOP

```

```

if [ -z "$RESPUESTA" ]; then
    echo '<p class="error">No se pudo obtener la blacklist de Banco 4 (servicio no disponible o
enlace caido).</p>'
else
    echo "<h2>Dominios bloqueados por Banco 4</h2>"
    echo "<ul>"
    echo "$RESPUESTA" | while read -r dominio; do
        [ -z "$dominio" ] && continue
        printf "<li>%s</li>\n" "$dominio"
    done
    echo "</ul>"
fi

```

```

cat << 'HTMLBOTTOM'
<p><a href="/cgi-bin/index.cgi">Volver al inicio</a></p>
</div>
</body>
</html>
HTMLBOTTOM
SCRIPT

```

```

chmod +x /var/www/localhost/cgi-bin/blacklist_banco4.cgi

```

```

cat > /var/www/localhost/cgi-bin/index.cgi << 'SCRIPT'
#!/bin/bash
echo "Content-type: text/html"
echo ""
DATA=/var/www/localhost/cgi-bin/data/cuentas.txt
cat << 'HTMLTOP'
<html>
<head><title>Banco 5 - Aplicacion Bancaria</title>
<style>`
body { font-family: Arial, sans-serif; background:#f2f2f2; margin:0; padding:20px; }
.header { background:#003366; color:white; padding:15px; border-radius:6px; }

```

```

.card { background:white; border-radius:6px; padding:15px; margin-top:15px; box-shadow:0
1px 3px rgba(0,0,0,0.2); }
table { width:100%; border-collapse:collapse; }
th, td { text-align:left; padding:8px; border-bottom:1px solid #ddd; }
input { padding:6px; margin:4px 0; width:100%; box-sizing:border-box; }
button, .btn { background:#003366; color:white; border:none; padding:10px 16px;
border-radius:4px; cursor:pointer; text-decoration:none; display:inline-block; }
</style>
</head>
<body>
<div class="header"><h1>Banco 5</h1><p>Servicio de Aplicacion Bancaria</p></div>
<div class="card">
<h2>Cuentas</h2>
<table>
<tr><th>No. Cuenta</th><th>Titular</th><th>Saldo</th></tr>
HTMLTOP
while IFS='|' read -r id nombre saldo; do
    printf "<tr><td>%s</td><td>%s</td><td>Q %.2f</td></tr>\n" "$id" "$nombre" "$saldo"
done < "$DATA"
cat << 'HTMLBOTTOM'
</table>
</div>
<div class="card">
<h2>Realizar Transferencia</h2>
<form method="post" action="/cgi-bin/transferir.cgi">
Cuenta origen:<br>
<input type="text" name="origen" placeholder="Ej. 1001"><br>
Cuenta destino:<br>
<input type="text" name="destino" placeholder="Ej. 1002"><br>
Monto:<br>
<input type="text" name="monto" placeholder="Ej. 100.00"><br><br>
<button type="submit">Transferir</button>
</form>
</div>
<div class="card">
<h2>Consulta Interbancaria</h2>
<p>Ver la lista de dominios bloqueados por Banco 4 (servicio publicado a traves del anillo
interbancario).</p>
<a class="btn" href="/cgi-bin/blacklist_banco4.cgi">Ver Blacklist de Banco 4</a>
</div>
</body>
</html>
HTMLBOTTOM
SCRIPT

```

```

chmod +x /var/www/localhost/cgi-bin/index.cgi

```

SERVIDOR 2 Y RRHH

```
mkdir -p /var/www/localhost/htdocs/archivos
```

```
echo "Reporte de nomina - Septiembre 2026" >
/var/www/localhost/htdocs/archivos/nomina_septiembre.txt
echo "Políticas de Recursos Humanos - Banco 5" >
/var/www/localhost/htdocs/archivos/politicas_rrhh.txt
echo "Solicitudes de vacaciones pendientes" >
/var/www/localhost/htdocs/archivos/vacaciones_pendientes.txt
```

```
cat > /etc/nginx/http.d/default.conf << 'EOF'
server {
    listen 80;
    server_name _;
    root /var/www/localhost/htdocs;
    autoindex on;
    autoindex_exact_size off;
    autoindex_localtime on;

    location / {
        try_files $uri $uri/ =404;
    }
}
EOF
```

```
rc-service nginx start
touch /etc/enable_nginx
```

```
pskill nginx
rc-service nginx start
```

inicio

```
cat > /var/www/localhost/generar_index.sh << 'EOF'
#!/bin/bash
DIR=/var/www/localhost/htdocs/archivos
OUT=/var/www/localhost/htdocs/index.html
```

```
cat > "$OUT" << 'HTMLTOP'
<html>
<head><title>Banco 5 - Servicio de Archivos</title>
<style>
```

```

body { font-family: Arial, sans-serif; background:#f2f2f2; margin:0; padding:20px; }
.header { background:#003366; color:white; padding:15px; border-radius:6px; }
.card { background:white; border-radius:6px; padding:15px; margin-top:15px; box-shadow:0
1px 3px rgba(0,0,0,0.2); }
ul { list-style:none; padding:0; margin:0; }
li { padding:12px 4px; border-bottom:1px solid #eee; display:flex;
justify-content:space-between; align-items:center; }
li:last-child { border-bottom:none; }
a { color:#003366; text-decoration:none; font-weight:bold; }
a:hover { text-decoration:underline; }
.meta { color:#888; font-size:0.85em; white-space:nowrap; margin-left:12px; }
.icon { margin-right:8px; }
</style>
</head>
<body>
<div class="header"><h1>Banco 5</h1><p>Servicio de Archivos - Recursos
Humanos</p></div>
<div class="card">
<h2>Archivos disponibles</h2>
<ul>

```

```

HTMLTOP

```

```

for f in "$DIR"/*; do
[ -f "$f" ] || continue
NOMBRE=$(basename "$f")
TAMANO=$(du -h "$f" | cut -f1)
FECHA=$(date -r "$f" "+%d/%m/%Y %H:%M")
printf '<li><span><span class="icon">\xf0\x9f\x93\x84</span><a
href="archivos/%s">%s</a></span><span class="meta">%s &middot; %s</span></li>\n' \
"$NOMBRE" "$NOMBRE" "$TAMANO" "$FECHA" >> "$OUT"
done

```

```

cat >> "$OUT" << 'HTMLBOTTOM'
</ul>
</div>
</body>
</html>
HTMLBOTTOM
EOF

```

```

chmod +x /var/www/localhost/generar_index.sh
/var/www/localhost/generar_index.sh

```



```
cat > /etc/nginx/http.d/default.conf << 'EOF'
```

```
server {
```

```
    listen 80;
```

```
    server_name _;
```

```
    root /var/www/localhost/htdocs;
```

```
    index index.html;
```

```
    location /archivos/ {
```

```
        autoindex off;
```

```
    }
```

```
    location / {
```

```
        try_files $uri $uri/ =404;
```

```
    }
```

```
}
```

```
EOF
```

```
rc-service nginx restart
```

API

```
mkdir -p /var/www/localhost/cgi-bin/api
```

```
cat > /var/www/localhost/cgi-bin/api/consulta.cgi << 'SCRIPT'
#!/bin/bash
echo "Content-type: application/json"
echo ""
```

```
DATA=/var/www/localhost/cgi-bin/data/cuentas.txt
```

```
# Extraer el parametro "cuenta" de la query string
CUENTA=$(echo "$QUERY_STRING" | tr '&' '\n' | grep '^cuenta=' | cut -d= -f2)
```

```
ENCONTRADO=0
NOMBRE=""
```

```
while IFS='|' read -r id nombre saldo; do
    if [ "$id" = "$CUENTA" ]; then
        ENCONTRADO=1
        NOMBRE=$nombre
    fi
done < "$DATA"
```

```
if [ -z "$CUENTA" ]; then
    echo '{"error":"falta el parametro cuenta"}'
elif [ "$ENCONTRADO" -eq 1 ]; then
    printf '{"cuenta":"%s","titular":"%s","existe":true}\n' "$CUENTA" "$NOMBRE"
else
    printf '{"cuenta":"%s","existe":false}\n' "$CUENTA"
fi
SCRIPT
```

```
chmod +x /var/www/localhost/cgi-bin/api/consulta.cgi
```

FIREWALL

```
cat > /etc/network/interfaces << 'EOF'
auto lo
iface lo inet loopback
```

```
auto eth0
iface eth0 inet static
    address 172.20.5.154
    netmask 255.255.255.252f
    gateway 172.20.5.153
```

```
auto eth1
iface eth1 inet manual
```

```
auto eth1.10
iface eth1.10 inet static
    address 172.20.5.1
    netmask 255.255.255.224
    vlan-raw-device eth1
```

```
auto eth1.20
iface eth1.20 inet static
    address 172.20.5.33
    netmask 255.255.255.224
    vlan-raw-device eth1
```

```
auto eth1.30
iface eth1.30 inet static
    address 172.20.5.65
    netmask 255.255.255.224
    vlan-raw-device eth1
```

```
auto eth1.40
iface eth1.40 inet static
    address 172.20.5.97
    netmask 255.255.255.224
    vlan-raw-device eth1
```

```
auto eth1.50
iface eth1.50 inet static
    address 172.20.5.129
    netmask 255.255.255.240
    vlan-raw-device eth1
```

```
EOF
```

```
echo "net.ipv4.ip_forward = 1" >> /etc/sysctl.conf
modprobe 8021q
ifup -a
sysctl -w net.ipv4.ip_forward=1
```

```
mkdir -p /etc/banco5
```

```
cat > /etc/banco5/blacklist_rrhh.txt << 'EOF'
facebook.com
instagram.com
tiktok.com
twitter.com
EOF
```

```
cat > /etc/banco5/blacklist_sistemas.txt << 'EOF'
netflix.com
youtube.com
twitch.tv
EOF
```

```
cat > /etc/banco5/apply_blacklist.sh << 'EOF'
#!/bin/sh
ipset create blacklist_rrhh hash:ip -exist
ipset create blacklist_sistemas hash:ip -exist
ipset flush blacklist_rrhh
ipset flush blacklist_sistemas
```

```
resolve_and_add() {
    set_name="$1"
    domain="$2"
    for ip in $(getent ahostsv4 "$domain" | awk '{print $1}' | sort -u); do
        ipset add "$set_name" "$ip" -exist
    done
}
```

```
while read -r domain; do
    [ -z "$domain" ] && continue
    resolve_and_add blacklist_rrhh "$domain"
done < /etc/banco5/blacklist_rrhh.txt
```

```
while read -r domain; do
    [ -z "$domain" ] && continue
    resolve_and_add blacklist_sistemas "$domain"
done < /etc/banco5/blacklist_sistemas.txt
EOF
```

```
chmod +x /etc/banco5/apply_blacklist.sh
echo "nameserver 8.8.8.8" > /etc/resolv.conf
/etc/banco5/apply_blacklist.sh
```

```
cat > /etc/firewall-banco5.sh << 'EOF'
#!/bin/sh
iptables -F
iptables -X
```

```
iptables -P INPUT DROP
iptables -P FORWARD DROP
iptables -P OUTPUT ACCEPT
```

```
iptables -A INPUT -i lo -j ACCEPT
```

```
iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
iptables -A FORWARD -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
```

```
iptables -A INPUT -i eth1.30 -p tcp --dport 22 -j ACCEPT
iptables -A INPUT -p icmp --icmp-type echo-request -j ACCEPT
```

```
iptables -A FORWARD -i eth1.10 -o eth1.50 -d 172.20.5.130 -p tcp --dport 80 -j ACCEPT
```

```
iptables -A FORWARD -i eth1.20 -o eth1.50 -d 172.20.5.131 -p tcp --dport 22 -j ACCEPT
iptables -A FORWARD -i eth1.20 -o eth1.50 -d 172.20.5.131 -p tcp --dport 80 -j ACCEPT
iptables -A FORWARD -i eth1.20 -o eth0 -m set --match-set blacklist_rrhh dst -j DROP
iptables -A FORWARD -i eth1.20 -o eth0 -j ACCEPT
```

```
iptables -A FORWARD -i eth1.30 -o eth1.50 -d 172.20.5.130 -p tcp -m multiport --dports
22,80 -j ACCEPT
iptables -A FORWARD -i eth1.30 -o eth1.50 -d 172.20.5.131 -p tcp --dport 22 -j ACCEPT
iptables -A FORWARD -i eth1.30 -o eth0 -m set --match-set blacklist_sistemas dst -j DROP
iptables -A FORWARD -i eth1.30 -o eth0 -j ACCEPT
```

```
iptables -A FORWARD -i eth1.40 -o eth1.50 -d 172.20.5.130 -p tcp --dport 80 -j ACCEPT
iptables -A FORWARD -i eth1.40 -o eth0 -j ACCEPT
```

```
# --- Servidor1 -> API de Banco 4 (excepcion puntual, consulta de blacklist) ---
iptables -A FORWARD -i eth1.50 -o eth0 -d 10.0.0.13 -p tcp --dport 8080 -j ACCEPT
EOF
```

```
chmod +x /etc/firewall-banco5.sh
/etc/firewall-banco5.sh
```

```
iptables -L FORWARD -v -n
iptables -L INPUT -v -n | head -3
```

SCRIPT PARA MOSTRAR EL BOTON PARA EL BANCO 4

```
cat > /var/www/localhost/cgi-bin/blacklist_banco4.cgi << 'SCRIPT'
#!/bin/bash
echo "Content-type: text/html"
echo ""
```

```
RESPUESTA=$(wget -qO- --timeout=5 http://10.0.0.13:8080/blacklist 2>/dev/null)
```

```
cat << 'HTMLTOP'
<html>
<head><title>Banco 5 - Blacklist de Banco 4</title>
<style>
body { font-family: Arial, sans-serif; background:#f2f2f2; margin:0; padding:20px; }
.header { background:#003366; color:white; padding:15px; border-radius:6px; }
.card { background:white; border-radius:6px; padding:15px; margin-top:15px;
box-shadow:0 1px 3px rgba(0,0,0,0.2); }
ul { padding-left:20px; }
li { padding:4px 0; }
a { color:#003366; }
.error { color:#a00; font-weight:bold; }
</style>
</head>
<body>
<div class="header"><h1>Banco 5</h1><p>Consulta interbancaria: Blacklist de
Banco 4</p></div>
<div class="card">
HTMLTOP
```

```
if [ -z "$RESPUESTA" ]; then
  echo '<p class="error">No se pudo obtener la blacklist de Banco 4 (servicio no
disponible o enlace caido).</p>'
else
  echo "<h2>Dominios bloqueados por Banco 4</h2>"
  echo "<ul>"
  echo "$RESPUESTA" | while read -r dominio; do
    [ -z "$dominio" ] && continue
    printf "<li>%s</li>\n" "$dominio"
  done
  echo "</ul>"
fi
```

```
cat << 'HTMLBOTTOM'
<p><a href="/cgi-bin/index.cgi">Volver al inicio</a></p>
</div>
</body>
</html>
HTMLBOTTOM
```

SCRIPT

```
chmod +x /var/www/localhost/cgi-bin/blacklist_banco4.cgi
```

show track brief